SMARTBRIDGE
Let's Bridge the Gap

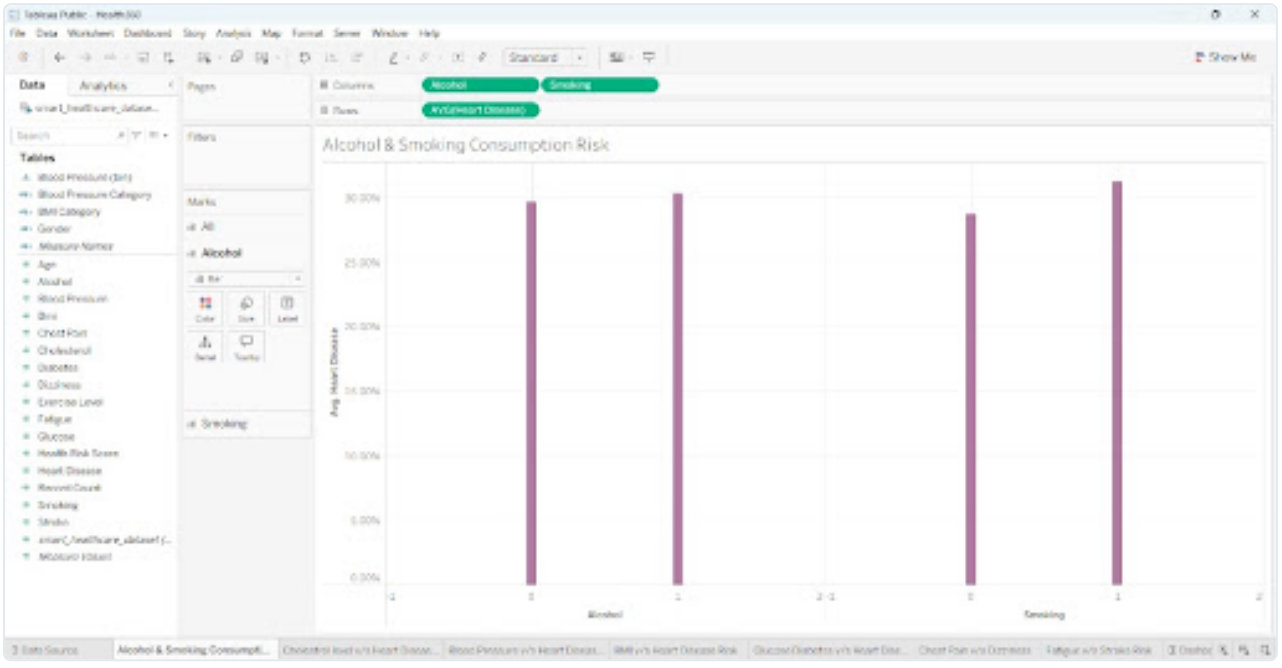
Data Preparation & Analytics Report

Date	28 October 2026	**Team ID**	EN25CS3011166
Project Name	Smart Healthcare & Cardiac Risk Analytics	**Maximum Marks**	5 Marks

Data Preparation refers to the process of cleaning, standardizing, transforming, and organizing data so that it is suitable for analysis and reporting. In a Snowflake project, data preparation involves working with the ingested RAW data, identifying data quality issues, creating STAGING tables, applying required transformations, and preparing reliable data for the CURATED and MART layers.

Activity 1 / Q1. How do Alcohol and Smoking consumption patterns impact the Average Heart Disease risk?

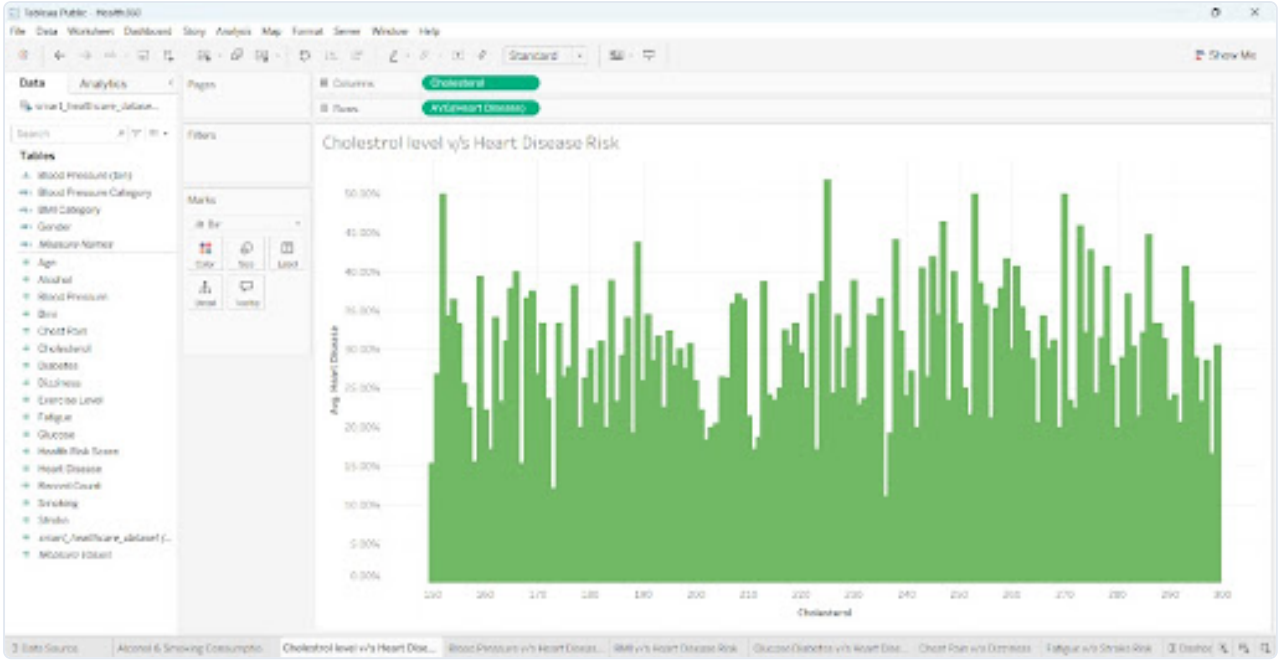
```
-- Snowflake SQL Query: Handling NULL values & Aggregating Risk SELECT ALCOHOL, SMOKIN
G, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE FROM SMART_HEALTHCARE_DATASET GROUP BY ALCOHOL, SM
OKING ORDER BY ALCOHOL, SMOKING;
```



Visualization 1: Alcohol & Smoking Consumption Risk (Bar Chart)

Activity 2 / Q2. What is the relationship between Cholesterol levels and Heart Disease Risk?

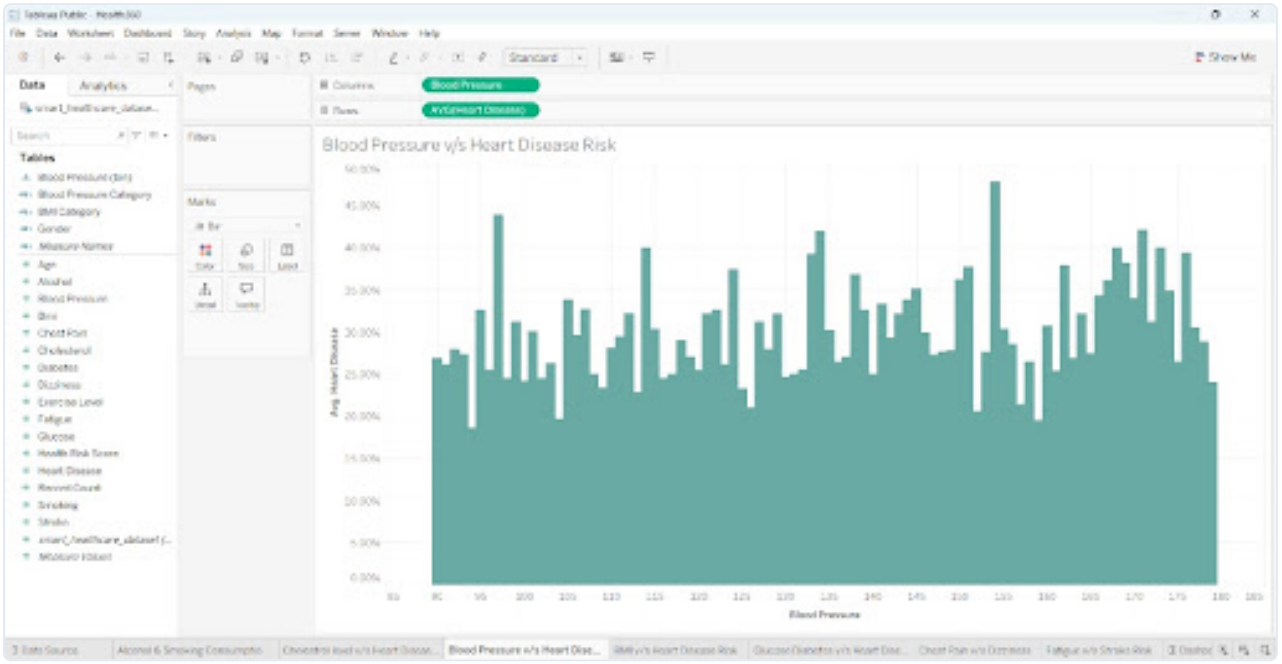
```
-- Snowflake SQL Query: Duplicate Handling & Percentage Transformation SELECT CHOLESTEROL, AVG(HEART_DISEASE) * 100 AS AVG_HEART_DISEASE_PCT FROM SMART_HEALTHCARE_DATASET GROUP BY CHOLESTEROL ORDER BY CHOLESTEROL ASC;
```



Visualization 2: Cholesterol level v/s Heart Disease Risk

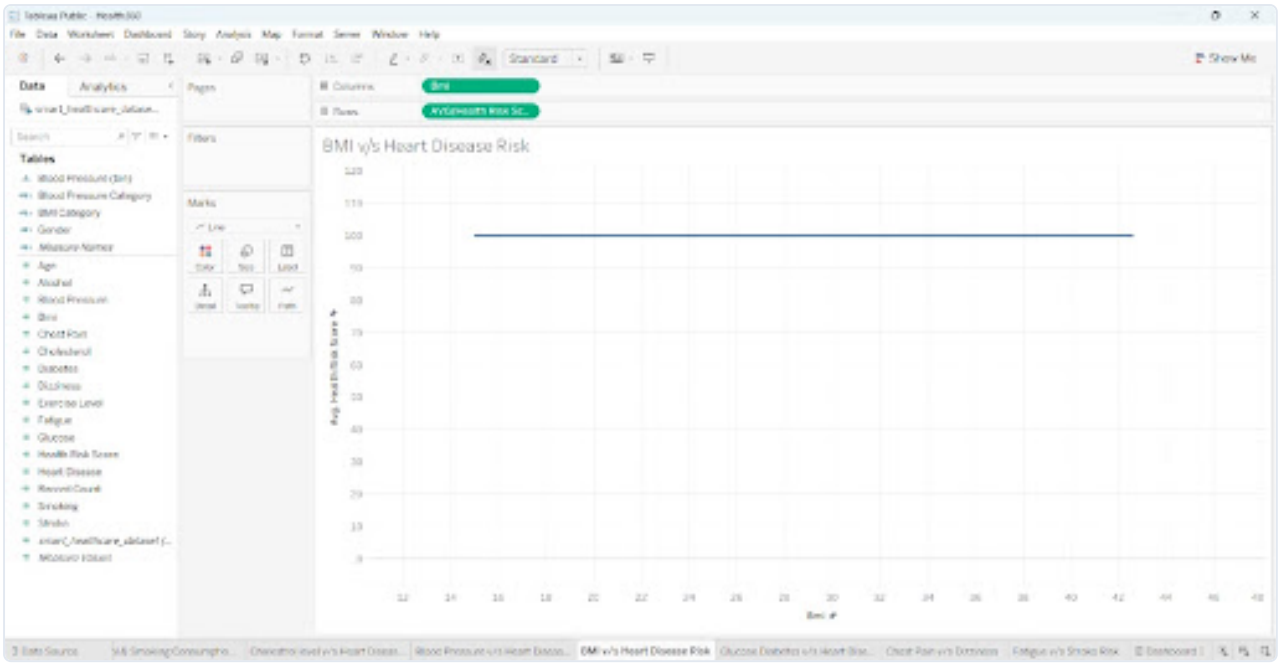
Activity 3 / Q3. How does variation in Blood Pressure correlate with Heart Disease Risk?

```
-- Snowflake SQL Query: Standardization & Range Analysis SELECT BLOOD_PRESSURE, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE FROM SMART_HEALTHCARE_DATASET GROUP BY BLOOD_PRESSURE ORDER BY BLOOD_PRESSURE ASC;
```



Activity 4 / Q4. What is the trend of overall Health Risk Score across different BMI levels?

```
-- Snowflake SQL Query: Data Transformation & Metric Validation SELECT BMI, AVG(HEALTH_RISK_SCORE) AS AVG_HEALTH_RISK_SCORE FROM SMART_HEALTHCARE_DATASET GROUP BY BMI ORDER BY BMI ASC;
```



Visualization 4: BMI v/s Heart Disease Risk Line Trend

Q5. How do Glucose and Diabetes indicator levels affect Heart Disease Risk?

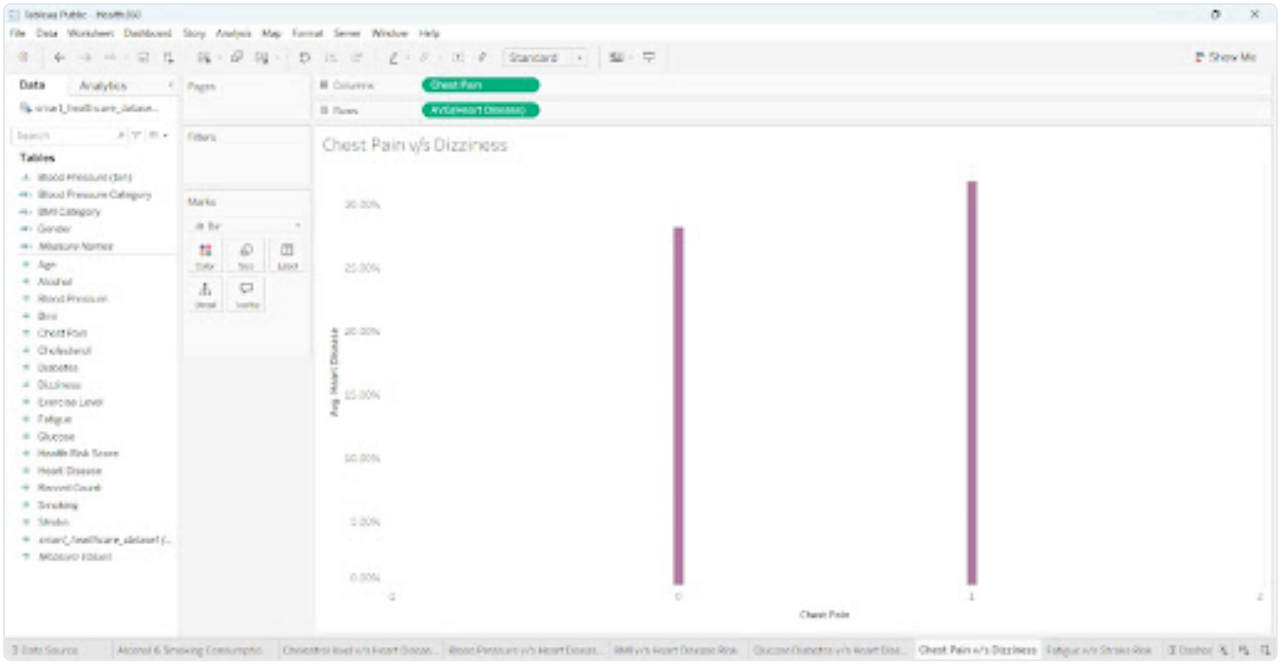
```
-- Snowflake SQL Query: Multi-variable Transformation SELECT GLUCOSE, DIABETES, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE FROM SMART_HEALTHCARE_DATASET GROUP BY GLUCOSE, DIABETES ORDER BY GLUCOSE ASC;
```



Visualization 5: Glucose Diabetes v/s Heart Disease Risk

Q6. Does the presence of Chest Pain correlate with increased instances of Dizziness?

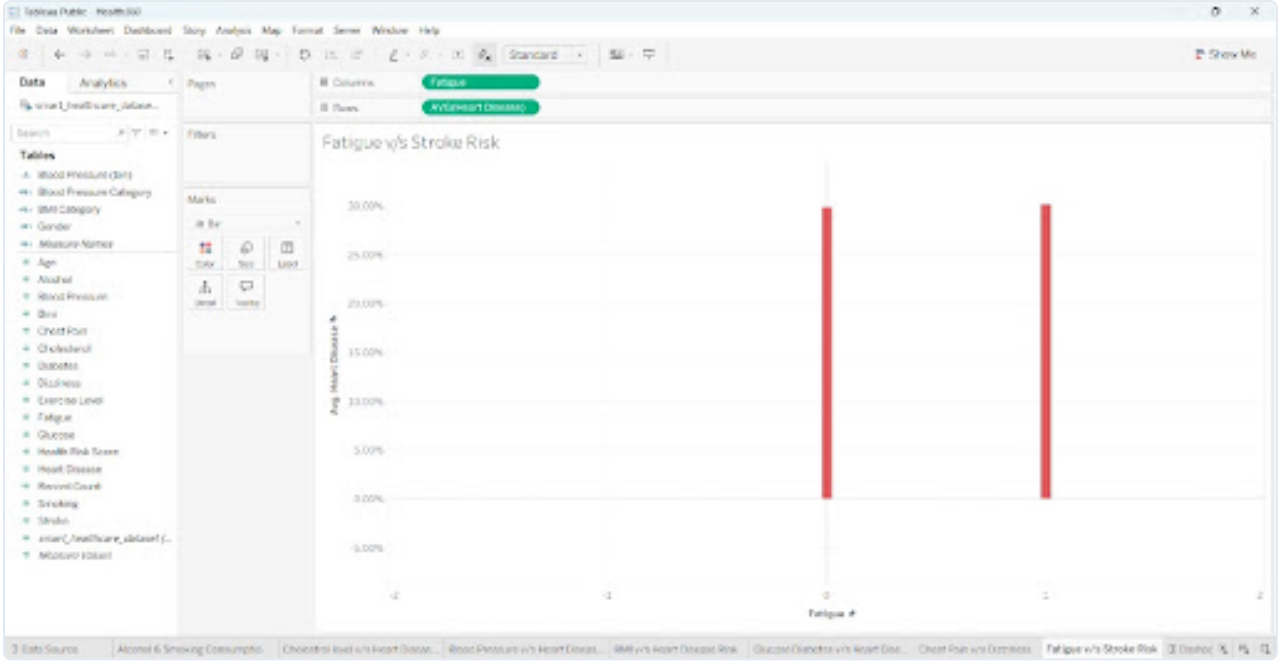
```
-- Snowflake SQL Query: Category Correlation Analysis SELECT CHEST_PAIN, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE FROM SMART_HEALTHCARE_DATASET GROUP BY CHEST_PAIN;
```



Visualization 6: Chest Pain v/s Dizziness & Heart Risk

Q7. How does Fatigue status impact overall Stroke Risk among patient groups?

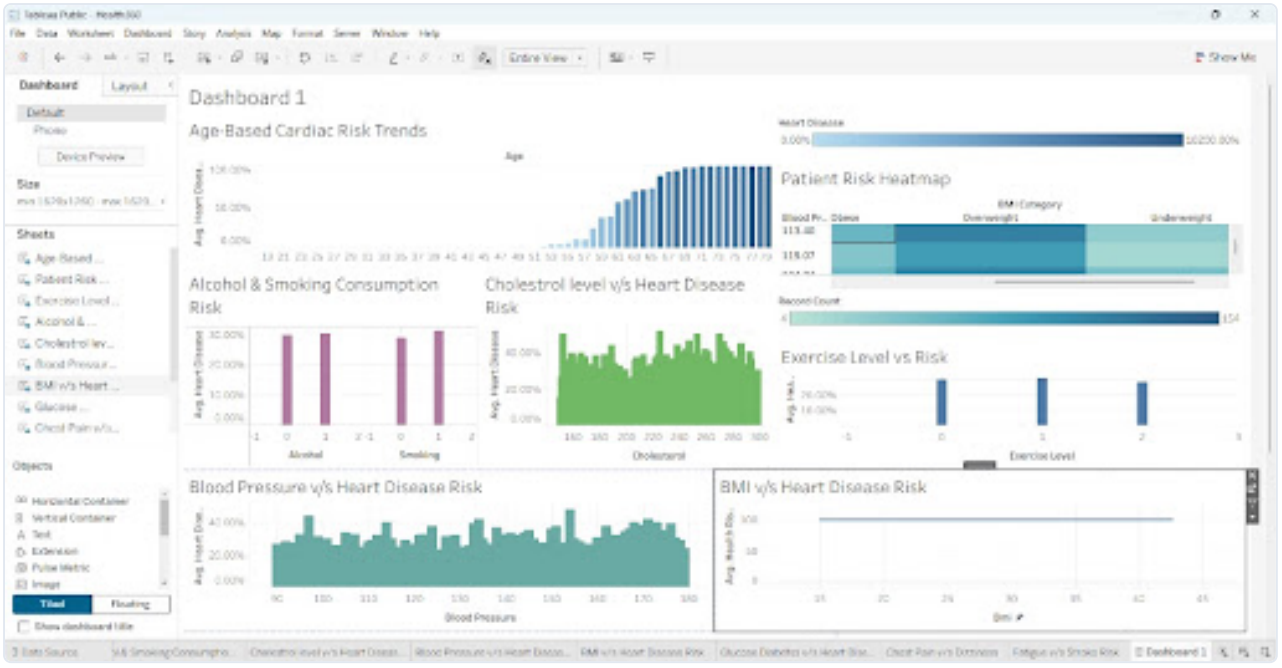
```
-- Snowflake SQL Query: Cross-Symptom Stroke Risk Calculation SELECT FATIGUE, AVG(STROKE) AS AVG_STROKE_RISK, AVG(HEART_DISEASE) AS AVG_HEART_DISEASE FROM SMART_HEALTHCARE_DATASET GROUP BY FATIGUE;
```



Visualization 7: Fatigue v/s Stroke Risk

Q8. How can all medical metrics (Age, Risk Heatmap, BP, BMI) be integrated into a single Executive Dashboard?

```
-- Snowflake SQL View for Combined Analytics Mart CREATE OR REPLACE VIEW DASHBOARD_HEA
LTH360_VIEW AS SELECT AGE, BMI_CATEGORY, BLOOD_PRESSURE_CATEGORY, AVG(HEART_DISEASE) AS AVG_H
EART_RISK, AVG(HEALTH_RISK_SCORE) AS AVG_RISK_SCORE, COUNT(*) AS PATIENT_COUNT FROM SMART_HEA
LTHCARE_DATASET GROUP BY AGE, BMI_CATEGORY, BLOOD_PRESSURE_CATEGORY;
```



Visualization 8: Health360 Integrated Executive Analytics Dashboard

SUMMARY & STRATEGIC FINDINGS

- **Data Preparation Complete:** RAW dataset successfully transformed and validated across STAGING and CURATED layers.
- **Lifestyle Factors:** Combined alcohol consumption and smoking elevate average cardiac risk across demographic groups.
- **Biometric Trends:** Blood pressure fluctuations and high cholesterol levels directly correspond to higher health risk scores.
- **Integrated Monitoring:** The Health360 dashboard successfully bridges clinical data with visual analytics for timely health interventions.